PHASE 1 : PowerQuery cleaning

1. Store_Master

- Loaded in PowerQuery and saw the column names where loaded as rows, therefore :
    - Trimmed Store_Name to removge spacing inconsistencies and standardized it
    - Also renamed the dataset to DimStore (naming it clearly for joining later)

2. Sales_Data
- Standardized Store_ID to text to align Store master
- Sales dates were in mixed formats, so they were standardized into a single Date type.
- Trimmed and clean for Store_Name to remove whitespace inconsistencies.
- Transaction_ID was not unique, so it was not used as a key so converted to text instead of int
- Missing and zero sales values were not imputed; instead, records were flagged as invalid for data quality tracking – cleanability
- **Sales dates were stored in mixed formats, so standardized date using a custom date column and removed old date**

    try Date.FromText([Date], "en-GB")

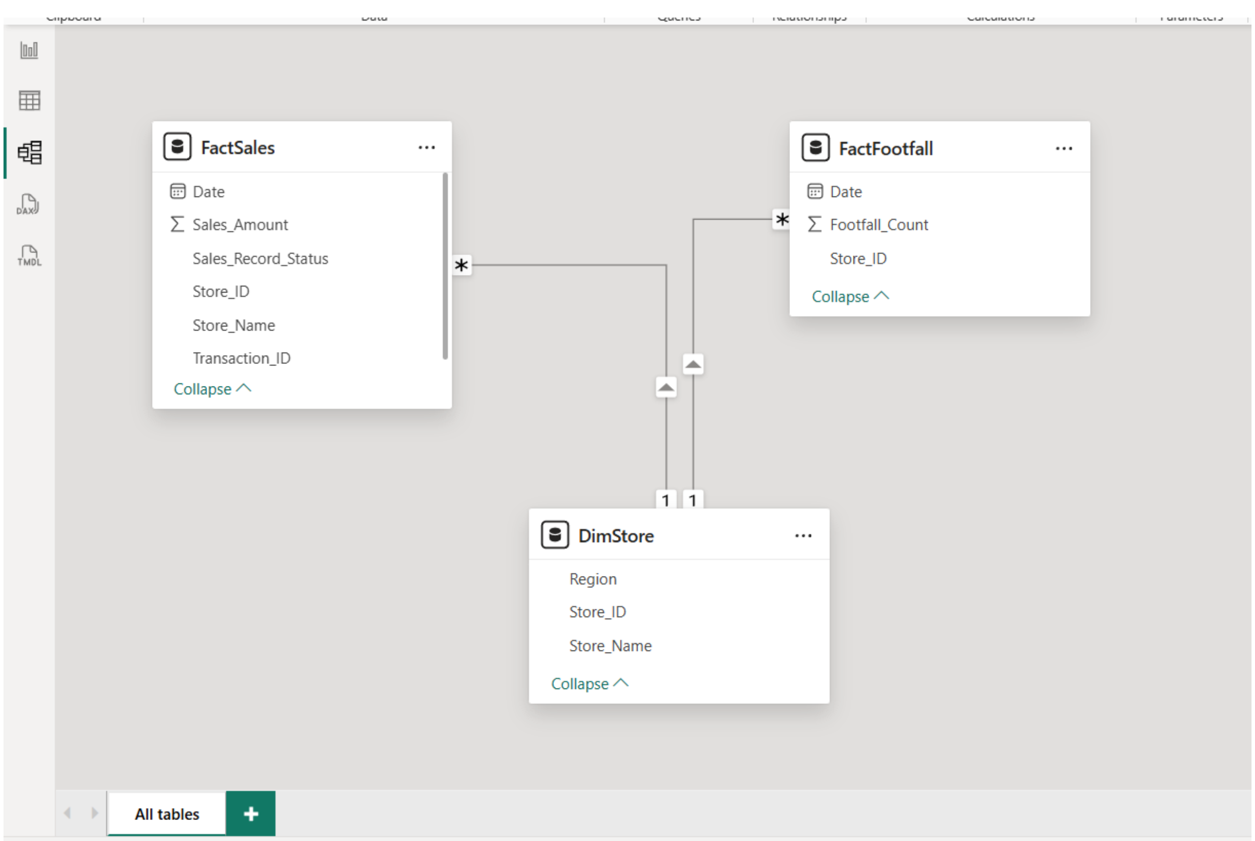
    otherwise try Date.FromText([Date], "en-US")

    otherwise null

- Changed the col name to FactSales

3. Footfall_Data
- Footfall data was standardized to Date and numeric format to match the other datasets column relation (better for analysis)
- Chnaged the col name to FactFootfall

PHASE 2 : Data Modeling



1. Created a **star schema** model using two fact tables and two shared dimensions:

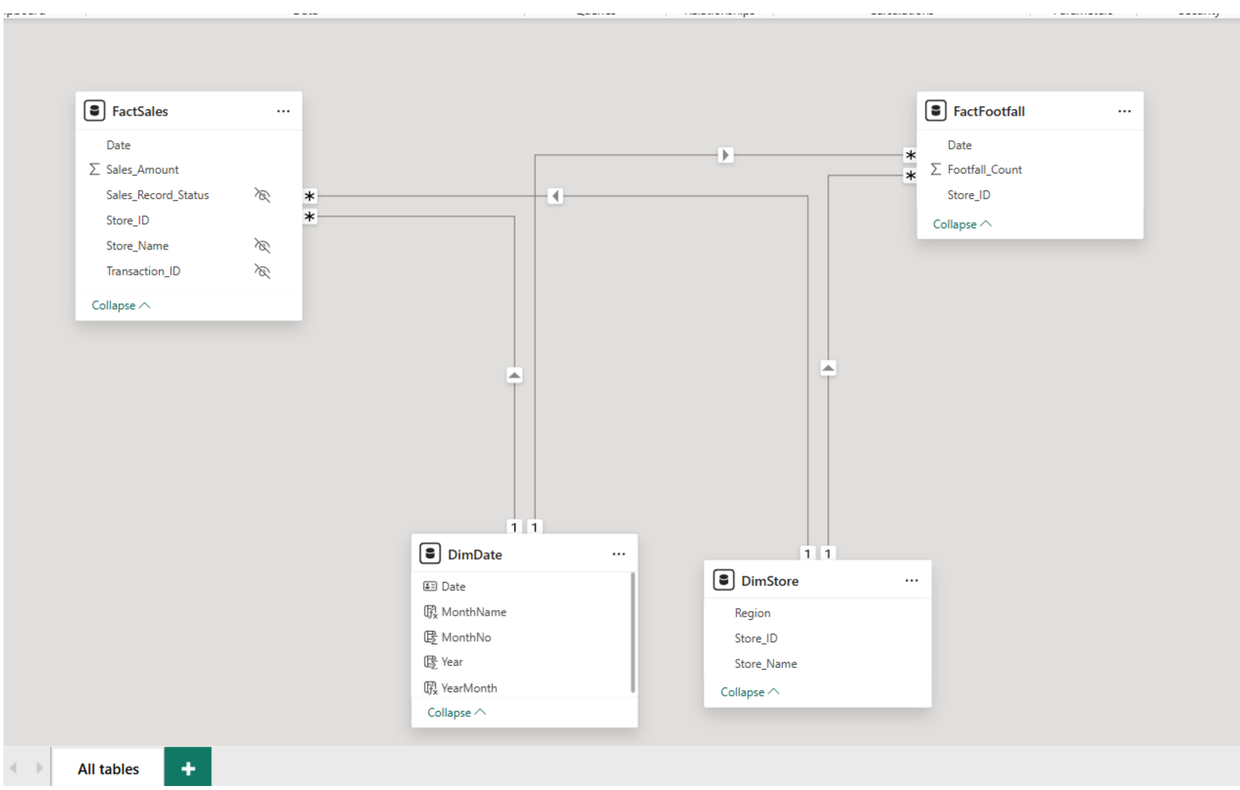
    - FactSales (transaction sales)
    - FactFootfall (daily store traffic)
    - DimStore (store + region mapping)
    - DimDate (calendar table)

    This was done because :

    - Sales and footfall are separate datasets, but both share the same business keys: **Store_ID** and **Date**.
    - Using shared dimensions ensures that slicers (date, store, region) filter both fact tables consistently.
    - A dedicated Date table was created to enable proper trend analysis and YoY comparisons.

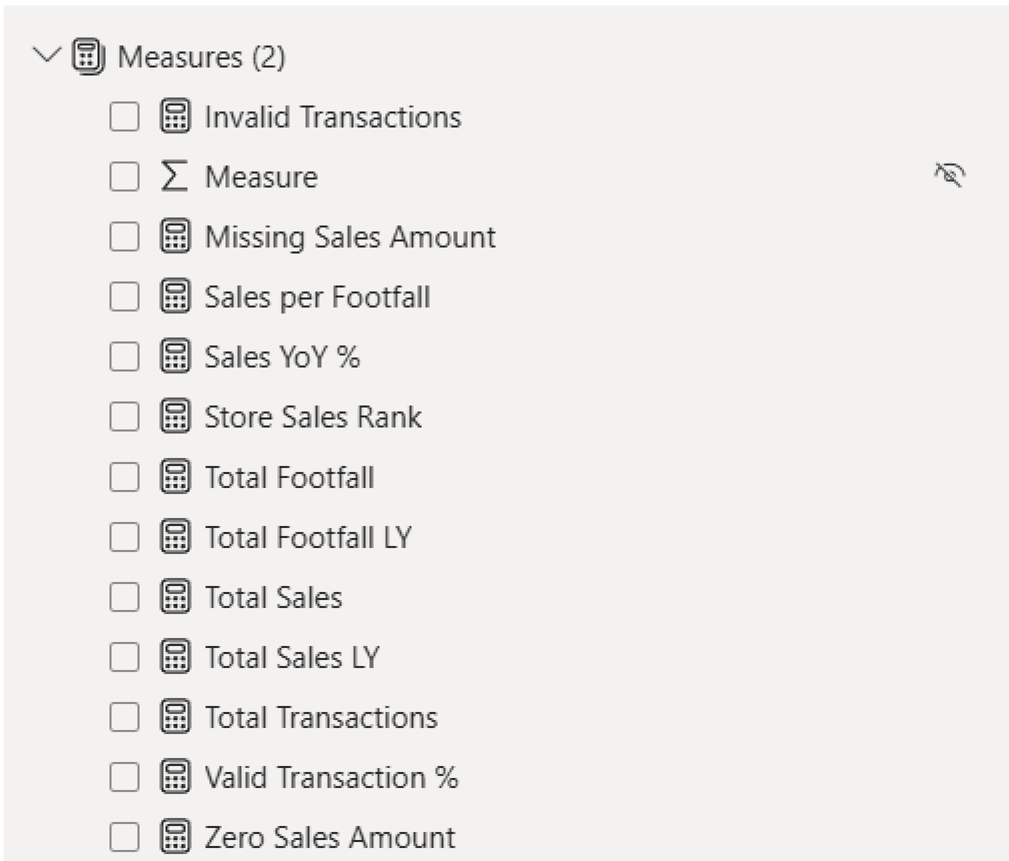
2. Custom Relationships created

- PowerBi already detected :
  - DimStore[Store_ID] - FactSales[Store_ID] (1-to-many)
  - DimStore[Store_ID] - FactFootfall[Store_ID] (1-to-many)
- Custom drag relationsips
  - DimDate[Date] - FactSales[Date] (1-to-many)
  - DimDate[Date] - FactFootfall[Date] (1-to-many)



PHASE 3 : DAX Custom Measures

- Created core KPIs for Total Sales, Total Footfall, and Sales per Footfall to evaluate performance and conversion.
- Created YoY measures using DimDate for current vs previous year comparison.
- Created data quality measures to track missing/zero sales records and calculate valid transaction percentage.



PHASE 4 : DASHBOARD

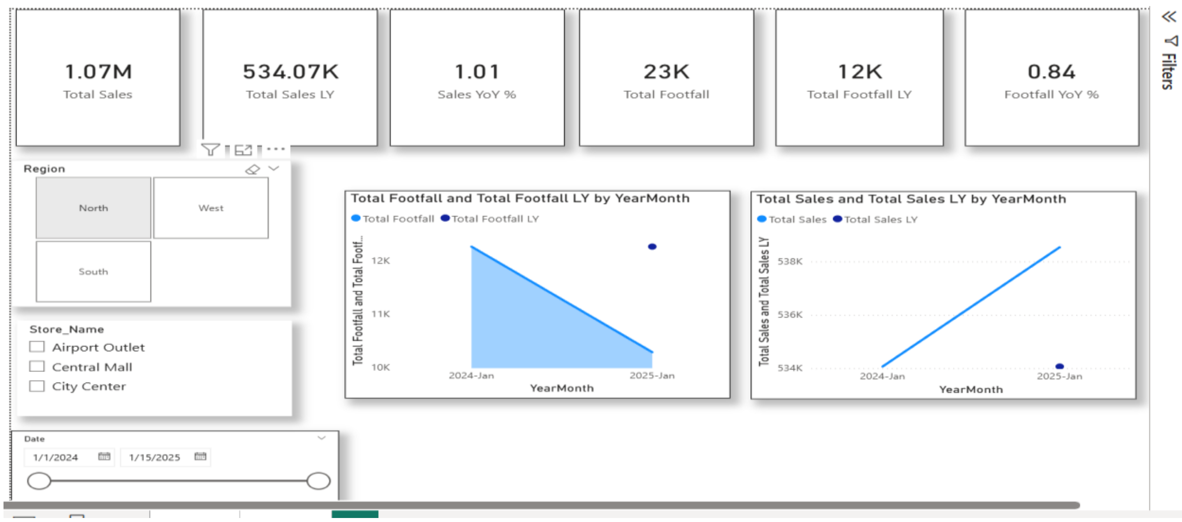
I plan to put 2 pages in dashboard to not make it messy

Page 1:

- Total Sales
- Total Footfall
- Sales vs Footfall Trend
- Store performance
- Data quality indicator (Valid Transaction %)
- Slicers for Date / Store / Region

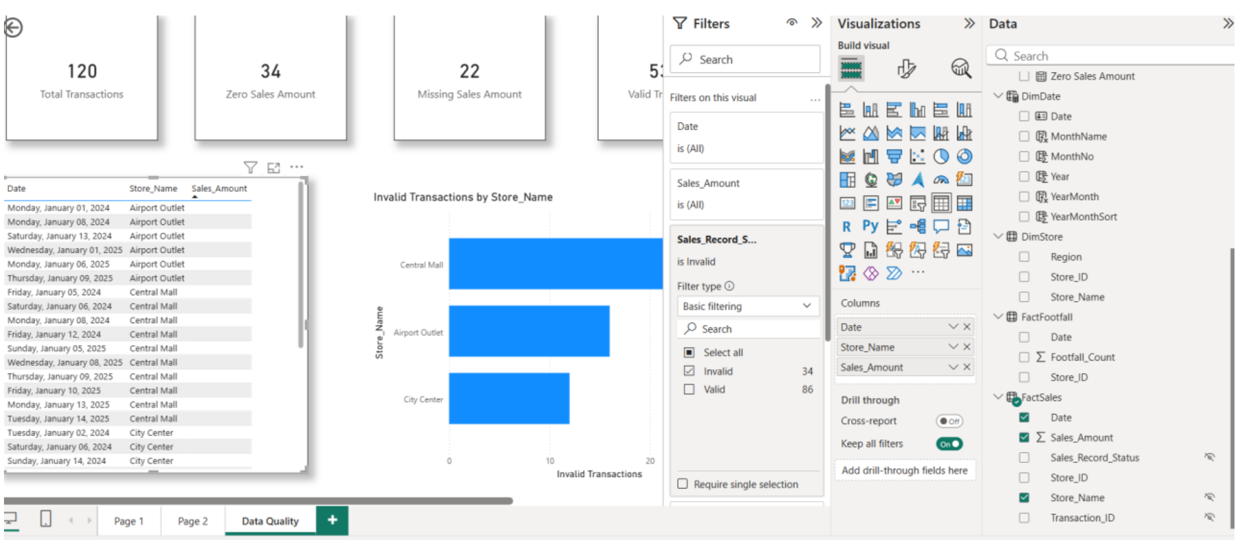
Page 2: YoY Comparisons (limited to Jan 2024 – Jan 2025) – 1 year Timeline

- Building a YoY page comparing current year vs previous year for both Sales and Footfall using time-intelligence measures.
- Including measures such as YoY growth % to quantify performance change.



Page 3: Data Quality Dashboard

- Mentioned in the assignment to evaluate quality measure
- Quality in the sense means nulls, inconsistency's
- Built a data quality page to track missing and zero sales records.
- Included invalid transaction counts and a detailed table to show affected records for transparency.



Page 4: Insights Representation

Page 5: Future recommendations & Author Information’s

PHASE 5 : DASHBOARD CSS

Sales-Footfall-Analytics-Dashboard-PowerBI.pdf

https://github.com/joelsiby02/Sales-Footfall-Analytics-Dashboard-PowerBI/blob/main/Dashboard/Sales-Footfall-Analytics-Dashboard-PowerBI.pdf